

08.-Docker-Compose-Build

Para esta actividad se desarrolla un stack de servicios relacionados utilizando **Docker Compose**, incorporando la creación de imágenes mediante **Dockerfile** y **context**.

Para ello se reutilizan las aplicaciones desarrolladas en el componente anterior:

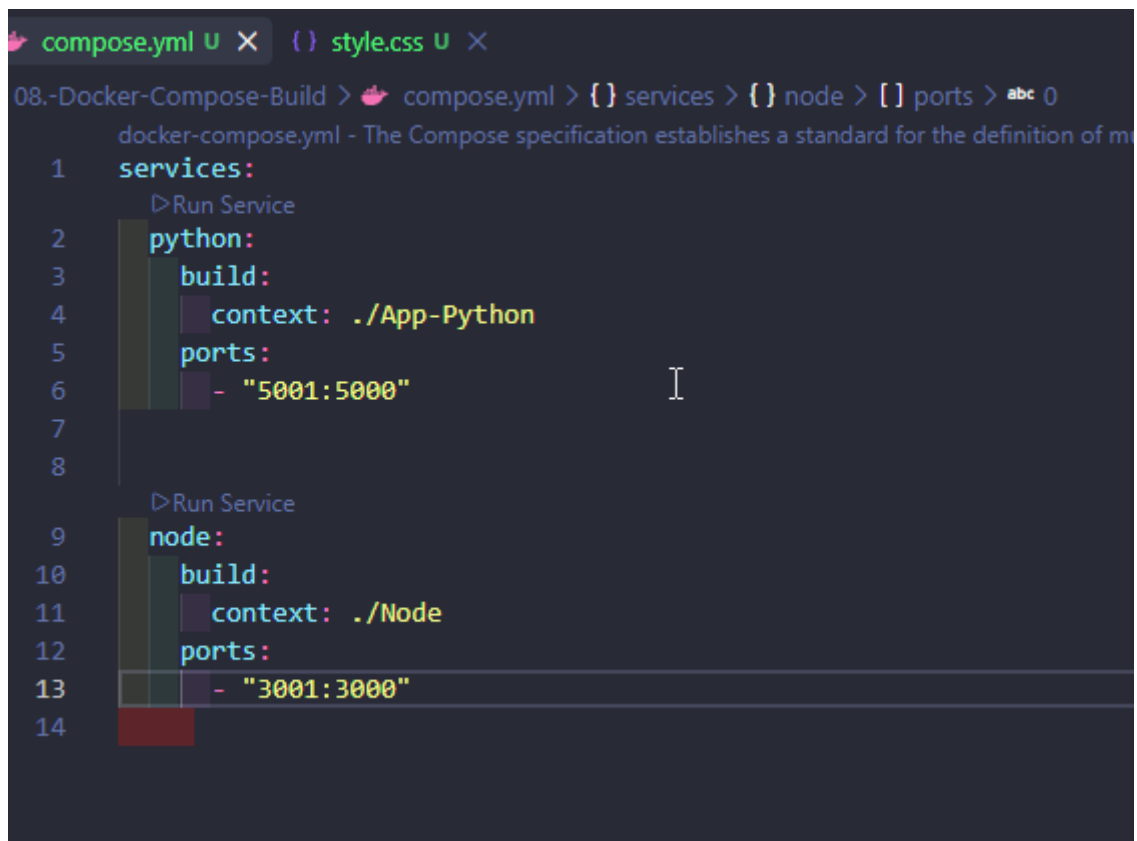
- **App-Python**
- **Node.js**

A diferencia del componente anterior, donde se trabajó principalmente con imágenes ya construidas, en esta actividad las imágenes son construidas directamente por Docker Compose a partir del código fuente y sus respectivos Dockerfile.

Configuración del Docker Compose

Creamos el archivo `compose.yml` utilizando `build` para indicar que las imágenes deben construirse a partir del código fuente, también definimos los puertos que usaremos.

`compose.yml`



```
compose.yml U X  {} style.css U X
08.-Docker-Compose-Build > {} compose.yml > {} services > {} node > [] ports > abc 0
docker-compose.yml - The Compose specification establishes a standard for the definition of m
1  services:
   ↳ Run Service
2  python:
3    build:
4      context: ./App-Python
5    ports:
6      - "5001:5000"
7
8
9  ↳ Run Service
10 node:
11   build:
12     context: ./Node
13   ports:
14     - "3001:3000"
```

Creación de imágenes

docker compose build

```
jezuz@DESKTOP-NQ62FUL:/mnt/c/Users/Jesus/Desktop/Galaxy Training/Docker/Examen-Final/08.-Docker-Compose-Build$ docker compose build
[+] Building 4.6s (30/30) FINISHED
=> [internal] load local bake definitions
=> => reading from stdin 1.25kB
=> [python internal] load build definition from Dockerfile
=> => transferring dockerfile: 659B
=> [node internal] load build definition from Dockerfile
=> => transferring dockerfile: 502B
=> [python] resolve image config for docker-image://docker.io/docker/dockerfile:1
=> [auth] docker/dockerfile:pull token for registry-1.docker.io
=> CACHED [node] docker-image://docker.io/docker/dockerfile:1@sha256:ecfaec9ed6d810b56388c508f4121597bfbba70d41a6dfeee4d8cad5f295fc32
=> => resolve docker.io/docker/dockerfile:1@sha256:ecfaec9ed6d810b56388c508f4121597bfbba70d41a6dfeee4d8cad5f295fc32
=> [python internal] load metadata for docker.io/library/python:3.12-slim
=> [node internal] load metadata for docker.io/library/node:20-alpine
=> [auth] library/python:pull token for registry-1.docker.io
=> [auth] library/node:pull token for registry-1.docker.io
=> [python internal] load .dockerignore
=> => transferring context: 244B
=> [python 1/6] FROM docker.io/library/python:3.12-slim@sha256:78387bc3881b8273120a12ebe6c1ab22b018cc2c9adf565ae1ac9b536e184ea
=> => exporting layers
=> => exporting manifest sha256:32e5703cbff70838f7c22e3cc1179fa0ff2696d4156e14a47af793414dae7539
=> => exporting config sha256:02c0a7419028de20bcc2f8e82ed67a3906a8acd421add20e2895940a405f8e0
=> => exporting attestation manifest sha256:097b697dee2f5939446a8861caf8331bb85c9241182e2c8ba672304833109bd1
=> => exporting manifest list sha256:8f75f8e755c664a2fdb30678097f35c539c57f5bb1240e089b47d88bdcf5688
=> => naming to docker.io/library/08-docker-compose-build-node:latest
=> => unpacking to docker.io/library/08-docker-compose-build-node:latest
=> [python] resolving provenance for metadata file
=> [node] resolving provenance for metadata file
[+] build 2/2
✔ Image 08-docker-compose-build-python Built
✔ Image 08-docker-compose-build-node Built
jezuz@DESKTOP-NQ62FUL:/mnt/c/Users/Jesus/Desktop/Galaxy Training/Docker/Examen-Final/08.-Docker-Compose-Build$
```

Levantamos los servicios y verificamos que estén activos

docker compose up -d

docker compose ps

```
✔ Image 08-docker-compose-build-node Built
jezuz@DESKTOP-NQ62FUL:/mnt/c/Users/Jesus/Desktop/Galaxy Training/Docker/Examen-Final/08.-Docker-Compose-Build$ docker compose up -d
[+] up 2/2
✔ Container 08-docker-compose-build-node-1 Started
✔ Container 08-docker-compose-build-python-1 Started
jezuz@DESKTOP-NQ62FUL:/mnt/c/Users/Jesus/Desktop/Galaxy Training/Docker/Examen-Final/08.-Docker-Compose-Build$ docker compose ps
```

NAME	IMAGE	COMMAND	SERVICE	CREATED
08-docker-compose-build-node-1	sha256:3457d851cf828161ab282536b5642393cccebd7709dd0ed2b2cefa0cace0bb7e	"docker-entrypoint.s..."	node	14 hours ago
08-docker-compose-build-python-1	sha256:dd50097cb1130018a6baff0597c34fcfe0b2037112b8c9e0b32d4eacca02d377	"gunicorn --bind 0.0..."	python	14 hours ago

Realizamos cambio en la hoja de estilos (**style.css**) y reconstruimos la imagen y luego recreamos el contenedor

docker compose build node

```
jezuz@DESKTOP-NQ62FUL:/mnt/c/Users/Jesus/Desktop/Galaxy Training/Docker/Examen-Final/08.-Docker-Compose-Build$ docker compose build node
[+] Building 7.2s (17/17) FINISHED
=> [internal] load local bake definitions
=> => reading from stdin 660B
=> [internal] load build definition from Dockerfile
=> => transferring dockerfile: 502B
```

docker compose up -d --force-recreate node

docker compose ps

```
✔ Image 08-docker-compose-build-node Built
jezuz@DESKTOP-NQ62FUL:/mnt/c/Users/Jesus/Desktop/Galaxy Training/Docker/Examen-Final/08.-Docker-Compose-Build$ docker compose up -d --force-recreate node
[+] up 1/1
✔ Container 08-docker-compose-build-node-1 Started
jezuz@DESKTOP-NQ62FUL:/mnt/c/Users/Jesus/Desktop/Galaxy Training/Docker/Examen-Final/08.-Docker-Compose-Build$ docker compose ps
```

NAME	IMAGE	COMMAND	SERVICE	CREATED	STATUS
08-docker-compose-build-node-1	08-docker-compose-build-node	"docker-entrypoint.s..."	node	About a minute ago	Up About a minute
08-docker-compose-build-python-1	sha256:dd50097cb1130018a6baff0597c34fcfe0b2037112b8c9e0b32d4eacca02d377	"gunicorn --bind 0.0..."	python	14 hours ago	Up 6 minutes